Functions

Outline

The concept of function

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The concept of function

Introduction: The concept of function

- In this section we introduce the concept of **function** in an intuitive way
- We start from concrete examples (a falling body, an investment) in which one quantity **depends** on another
- We will understand that what characterizes a function is the **uniqueness** of the relation between input and output

Variable quantities

The existence of a **variable quantity** implies the existence of a **relation** between **two quantities**, that is, the dependence of one quantity on another.

- In each case, this relation follows a certain **law** or **formula**.

Example (1/2): laws/formulas $\rightarrow$ functions

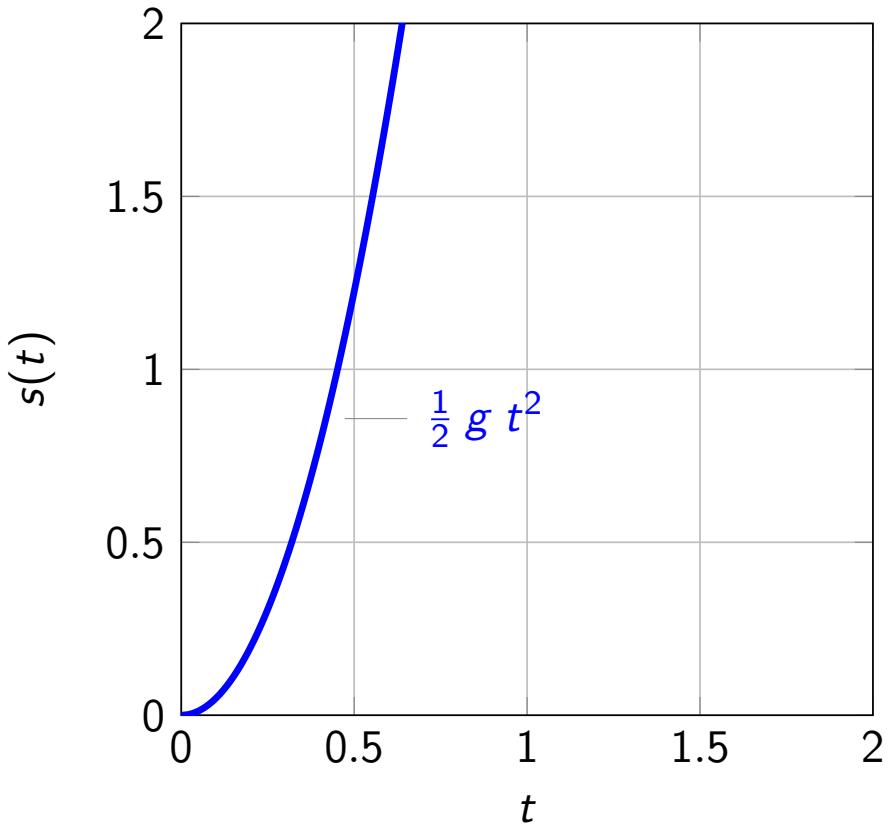
If a heavy object is dropped from a certain height, the distance traveled by the object varies with time t according to the formula:

$$s(t) = \frac{1}{2}\, g\, t^2 \qquad \text{with} \quad t \geq 0$$

where $g \approx 9.8$ is the gravitational acceleration constant.

Hence, the time t is associated with the distance traveled $s(t)$: $t \mapsto s(t)$.

Example (2/2): laws/formulas $\rightarrow$ functions



Example (1/3): laws/formulas $\to$ functions

If a unit capital is invested for one year at the annual rate i, with interest paid monthly at a rate $\frac{i}{12}$, we have:

- after the first month, a capital equal to:

$$1 + \frac{i}{12} \cdot 1 = \underbrace{1 + \frac{i}{12}}_{\alpha}$$

- after the second month, a capital equal to:

$$\underbrace{1 + \frac{i}{12}}_{\alpha} + \frac{i}{12} \underbrace{\left(1 + \frac{i}{12}\right)}_{\alpha} = \left(1 + \frac{i}{12}\right)\left(1 + \frac{i}{12}\right) = \underbrace{\left(1 + \frac{i}{12}\right)^2}_{\beta}$$

Example (2/3): laws/formulas $\to$ functions

- after the third month, a capital equal to:

$$\underbrace{\left(1+\frac{i}{12}\right)^2}_{\beta}+\frac{i}{12}\underbrace{\left(1+\frac{i}{12}\right)^2}_{\beta}=\left(1+\frac{i}{12}\right)^2\left(1+\frac{i}{12}\right)$$

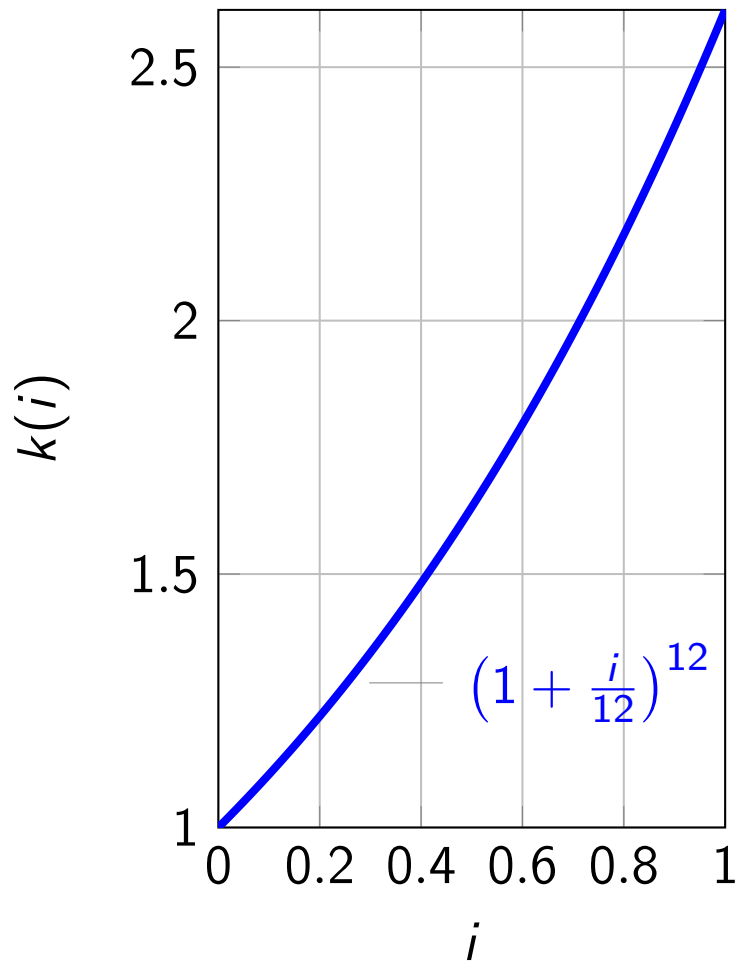
$$=\left(1+\frac{i}{12}\right)^3$$

Repeating the argument, the capital at the end of the year therefore depends on the rate i according to the formula:

$$k(i)=\left(1+\frac{i}{12}\right)^{12} \qquad \text{with} \quad i\in[0,1]$$

Hence, the rate i is associated with the capital $k(i)$ at the end of the year: $i \mapsto k(i)$.

Example (3/3): laws/formulas $\rightarrow$ functions



Inputs and outputs

- In each of the previous examples, a *real number* (**input**) is associated ***uniquely*** with another *real number* (**output**). It is precisely the **uniqueness of the relation** that characterizes a function.
- In general, the **admissible inputs** for a given relation (function) are subject to natural restrictions, related to the very nature of the relation.

Example: admissible inputs

- In the first example the "starting" real number has the meaning of *time*; imagining that the object is dropped at an initial time $t = 0$, it is clear that we must restrict ourselves to times $t \geq 0$.
- In the second example, clearly we must have $0 \leq i \leq 1$, where 1 corresponds to a rate of 100%.

Summary

The concept of function:

- A variable quantity expresses the **dependence** of one quantity on another, according to a **law** or **formula**
- Examples: the distance $s(t) = \frac{1}{2}gt^2$ traveled by a falling body, the capital $k(i) = \left(1 + \frac{i}{12}\right)^{12}$
- What characterizes a function is **uniqueness**: each input corresponds to exactly one output
- The **admissible inputs** are subject to natural restrictions

Definition of function, domain, codomain and image

Introduction: Definition of function, domain, codomain and image

- In this section we give the **formal definition** of a function between any two sets
- We introduce the concepts of **domain**, **codomain** and **image**, and the notation $f : A \to B$, $x \mapsto f(x)$
- This language will be used throughout the rest of the course

Domain

Definition 1: of domain

The set of admissible inputs for a given function is called the ***domain****.*

- The expressions **independent variable** (for a generic *input*) and **dependent variable** (for the *output*) are often used.

Example: other types of relations

Let us consider the set of students of a given course.

- If A is the set of students and B is the set of their names, the relation that associates each student with his or her name is *unique* (single-valued).
- In this case neither A nor B is a set of numbers, but nevertheless a unique (single-valued) relation between these two sets is well defined.

On the other hand, the converse is not necessarily true: there could be two students with the same name.

Function

Definition 2: of function

Given any two sets A, B, a ***function*** *f with* ***domain*** *A and values in B (or also "with* ***codomain*** *B") is any law that associates with each element of A one and only one element of B.*

Notation

- The notation:
$$f : A \to B$$
(which reads "f defined from A to B"), indicates the **domain** and the **codomain** of the function f.
- The notation
$$f : x \mapsto f(x)$$
(which reads "f maps x to $f(x)$") indicates how the function f acts on the elements.
- The **symbol $f(x)$ denotes the output or the value** that the function f associates with x, and it must not be confused with the symbol f, which denotes the **function itself**.

Example: notation for functions

For the function

$$k(i) = \left(1 + \frac{i}{12}\right)^{12}$$

of the previous example we would have:

$$k : [0, 1] \to \mathbb{R}$$

$$k : i \mapsto \left(1 + \frac{i}{12}\right)^{12}$$

Equality of two functions

Definition 3: of equality of two functions

*Two functions f and g are **equal** if they have the same domain A, the same codomain B and if*

$$f(a) = g(a), \qquad \text{for every} \quad a \in A.$$

- A function is not just a "formula": **the domain and the codomain are part of the function**

Example: equal functions and different functions

Consider the three functions

$$f : \mathbb{R} \to \mathbb{R}, \quad f : x \mapsto x^2$$

$$g : \mathbb{R} \to [0, +\infty), \quad g : x \mapsto x^2$$

$$h : [0, +\infty) \to \mathbb{R}, \quad h : x \mapsto x^2$$

- $f \neq g$: same domain and same law, but different codomains
- $f \neq h$: same codomain and same law, but different domains

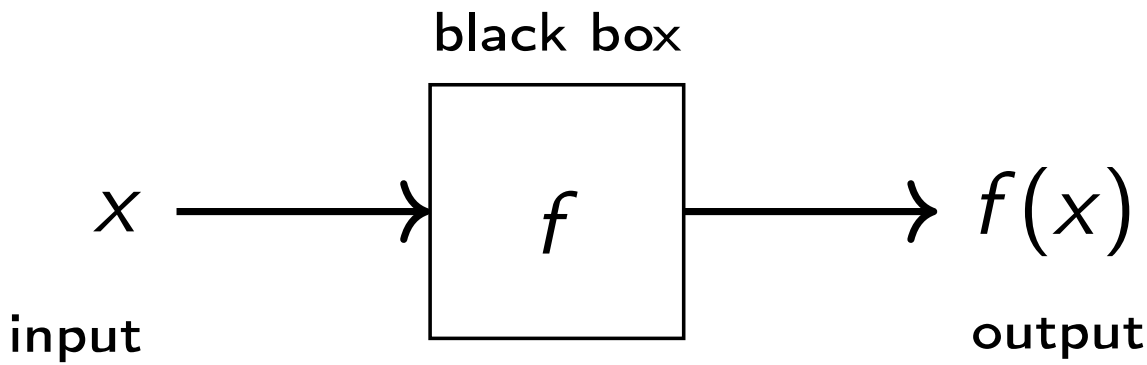
The notation $y = f(x)$

- The notation
$$y = f(x)$$
is also used to indicate that the variable y is a function of x. In this case y is the output or the value that f associates with the input x.

For brevity, sometimes the value of $f(x)$ is defined directly in terms of x, as for example $f(x) = x^2 + 10$.

The function as a black box

- In general, we can think of a function as a **black box** that associates with each admissible input x (**input**) a unique output $f(x)$ (**output**):



Image

Definition 4: of image and image of the domain

The output corresponding to x is called the ***image*** *of x; the set of possible outputs is called the* ***image of the domain*** A ***under*** f *and is denoted by the symbol $f(A)$ or* $\operatorname{Im} f$.

- Note that in the notation $f : A \to B$, the codomain B can be larger than the image $f(A)$, that is, in general we have

$$f(A) \subseteq B.$$

- If f is real-valued, we usually write $f : A \to \mathbb{R}$ without specifying the actual image of f.

Image of a subset of the domain

Definition 5: of image of a subset of the domain

Given a function $f : A \to B$ *and a subset* $A' \subseteq A$*, the* ***image of*** A' ***under*** f *is*

$$f(A') = \{b \in B : \quad b = f(a) \ \text{ for some } \ a \in A'\}.$$

- Taking $A' = A$ we recover the image of the domain (the *range* of f):

$$f(A) = \{b \in B : \quad b = f(a) \ \text{ for some } \ a \in A\} \subseteq B$$

Example: image of the domain and of a subset

Consider the function $f : \mathbb{N} \to \mathbb{N}$, $f : n \mapsto 2\,n$.

- The image of its domain is

$$f(\mathbb{N}) = \{m \in \mathbb{N} : \quad m = 2\,n \ \text{ for some } \ n \in \mathbb{N}\},$$

  that is, the set of the **nonnegative even numbers**: in particular $f(\mathbb{N}) \subsetneq \mathbb{N}$

- The image of the subset $A' = \{0, 1, 2, 3\}$ is $f(A') = \{0, 2, 4, 6\}$

Summary

Definition of function:

- The **domain** is the set of admissible inputs
- A **function** $f : A \to B$ associates with each element of A *one and only one* element of B (**codomain**)
- $f(x)$ is the value associated with x, f is the function itself; we also write $y = f(x)$

Image:

- $f(A) = \operatorname{Im} f$ is the set of possible outputs and in general $f(A) \subseteq B$

Surjections, injections and bijections

Introduction: Surjections, injections and bijections

- In this section we classify functions according to how they relate the domain and the codomain
- We introduce **surjective**, **injective** and **bijective** functions, with their set representation
- We prove the relations between the cardinalities of domain and codomain for finite sets: these concepts will be fundamental for inverse functions

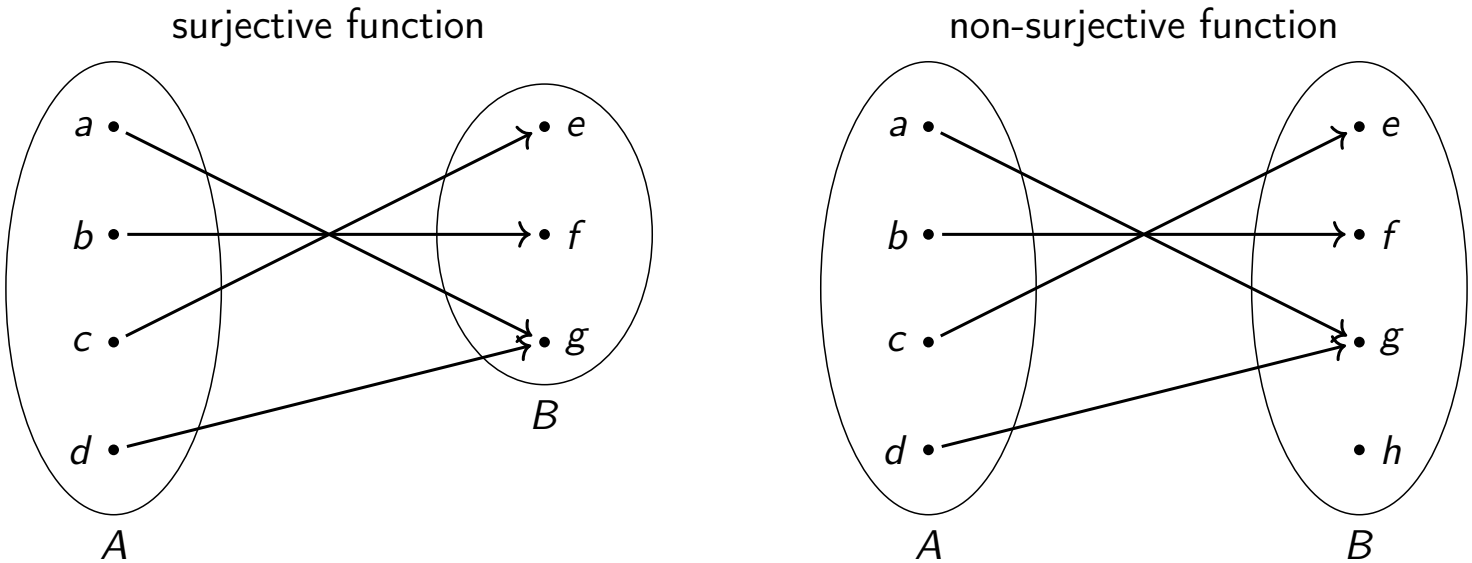
Surjection

Definition 6: of surjection (surjective function)

*A function f is a **surjection** if the image of its domain coincides with its codomain.*

- If $f : A \to B$ is a surjection, we also say that f **maps** A **onto** B
- The word *mapping* denotes an arbitrary function: it is the preposition *onto* that expresses surjectivity

Surjection: set representation



Example: surjective and non-surjective functions

- The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since every element of the codomain $\mathbb{N}$ is the image of some value of the domain.
- The function $f(n) = 2\,n$ is not a surjective function from $\mathbb{N}$ to $\mathbb{N}$, since no argument of f produces 3 as a value.
- The function $f(n) = 2\,n$ is, however, a surjective function from the natural numbers to the even numbers.

Surjections and cardinality

Observation 1

Given two ***finite*** *sets A and B, and a function $f : A \to B$, if f is surjective then $|A| \geq |B|$.*

Proof (1/2)

Write $B = \{b_1, b_2, \ldots, b_m\}$, with $m = |B|$, and for every $i \in \{1, 2, \ldots, m\}$ consider the set of the inputs whose output is b_i:

$$A_i = \{a \in A : \ f(a) = b_i\} \subseteq A.$$

- Every A_i is **nonempty**: f is surjective, hence $|A_i| \geq 1$
- The sets $A_1, A_2, \ldots, A_m$ are **pairwise disjoint**: if $a \in A_i \cap A_j$ then $b_i = f(a) = b_j$, because f associates with a one and only one output
- Their **union is the whole of** A: every $a \in A$ lies in A_i with $b_i = f(a)$

Proof (2/2)

The sets $A_1, A_2, \ldots, A_m$ thus form a *partition* of A and, since A is finite, we count the elements of A by summing the cardinalities of the blocks:

$$|A| = \sum_{i=1}^{m} |A_i| \geq \sum_{i=1}^{m} 1 = m = |B|$$

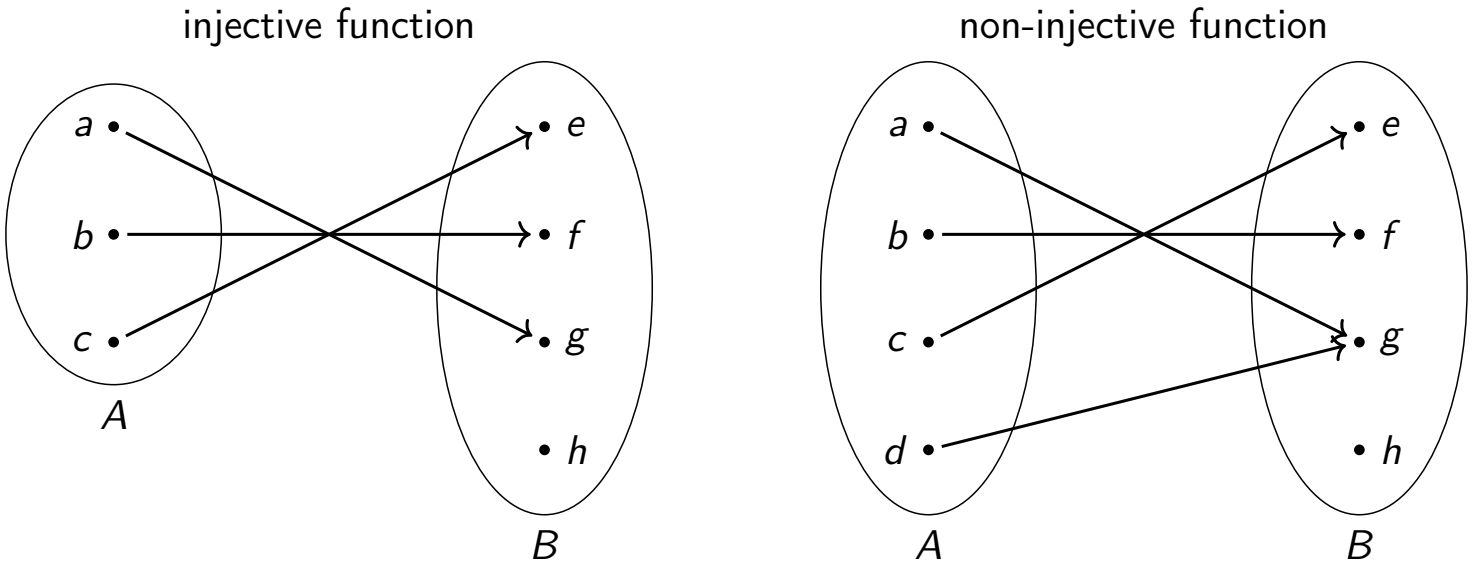
□

Injection

Definition 7: of injection (injective function)

*A function f is an **injection** if distinct arguments of f produce distinct values, that is, if $a \neq b$ implies $f(a) \neq f(b)$.*

Injection: set representation



Example

- The function $f(n) = 2\,n$ is an injective function from $\mathbb{N}$ to $\mathbb{N}$, since each even number b is the image under f of exactly one element of the domain, namely $n = \frac{b}{2}$.
- The function $f(n) = \lfloor \frac{n}{2} \rfloor$ is not an injective function, since the value 1 is obtained from two arguments: 2 and 3.

Injections and cardinality

- An injection is also called a **one-to-one** function.

Observation 2

*Given two **finite** sets A and B, and a function $f : A \to B$, if f is injective then $|A| \leq |B|$.*

Proof (1/2)

Write $A = \{a_1, a_2, \ldots, a_n\}$, with $n = |A|$, and consider the n outputs

$$f(a_1),\ f(a_2),\ \ldots,\ f(a_n) \in f(A).$$

- These outputs are **all distinct**: if $i \neq j$ then $a_i \neq a_j$ and, since f is injective, $f(a_i) \neq f(a_j)$
- Every element of $f(A)$ **appears** in the list: every $b \in f(A)$ has the form $b = f(a)$ with $a \in A$, that is, $a = a_i$ for some i

Proof (2/2)

The list $f(a_1), f(a_2), \ldots, f(a_n)$ therefore enumerates the elements of $f(A)$ without repetitions:

$$|A| = n = |f(A)|.$$

Finally $f(A) \subseteq B$ and B is finite, hence $|f(A)| \leq |B|$. Putting the two relations together:

$$|A| \leq |B|.$$

□

Warning: finite sets

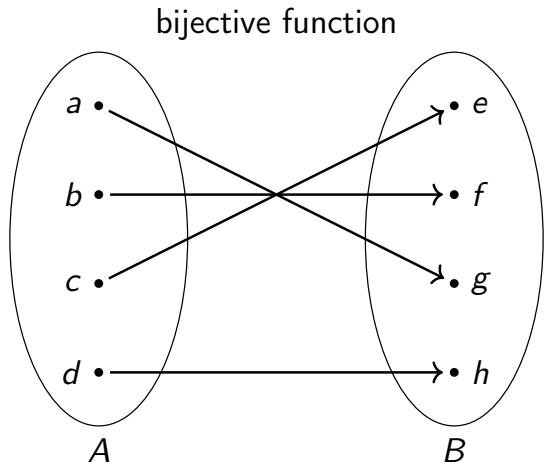
- The two remarks are stated for **finite** sets: the proofs *count* elements, and counting only makes sense for finite sets
- Comparing the "sizes" of two infinite sets requires a different notion of cardinality, introduced in the chapter "Cardinality of infinite sets" of Part 2

Bijection

Definition 8: of bijection (bijective function or one-to-one correspondence)

*A function f is a **bijection** if: (i) it is injective and (ii) it is surjective.*

- Set representation:



Example

- The function $f(n) = (-1)^n \left\lceil \frac{n}{2} \right\rceil$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$. The values of the function are:

  $$f(0) = 0, \quad f(1) = -1, \quad f(2) = 1, \quad f(3) = -2, \quad f(4) = 2 \ldots$$

- The function is injective, since no element of $\mathbb{Z}$ is the image of more than one element of $\mathbb{N}$.
- The function is surjective, since all elements of $\mathbb{Z}$ are images of some element of $\mathbb{N}$.

Bijections and permutations

- A bijection is also called a **one-to-one** correspondence, since it pairs elements of the domain with elements of the codomain.
- A bijection from a set A to itself is also called a **permutation**.

Summary

Surjections, injections and bijections:

- **Surjective**: the image coincides with the codomain; if f is surjective then $|A| \geq |B|$
- **Injective**: $a \neq b \Rightarrow f(a) \neq f(b)$; if f is injective then $|A| \leq |B|$
- **Bijective**: injective and surjective; a bijection from A to itself is a **permutation**
- Example: $f(n) = (-1)^n \lceil \frac{n}{2} \rceil$ is a bijection from $\mathbb{N}$ to $\mathbb{Z}$

Inverse of a bijection

Introduction: Inverse of a bijection

- A bijection pairs every element of the domain with an element of the codomain and vice versa
- We see how to follow this pairing in the opposite direction, defining the **inverse function**
- This is the idea behind the inverse functions of functions of a real variable

Inverse of a bijection

Definition 9: of inverse of a bijection

*Given a bijection $f : A \to B$, its **inverse** is the function*

$$f^{-1} : B \to A, \qquad f^{-1}(b) = a \iff f(a) = b.$$

- In other words: $f^{-1}\big(f(a)\big) = a$ for every $a \in A$ and $f\big(f^{-1}(b)\big) = b$ for every $b \in B$

Why the definition is well posed

Given $b \in B$:

- the **surjectivity** of f guarantees that there exists *at least one* $a \in A$ with $f(a) = b$
- the **injectivity** of f guarantees that there exists *at most one*, because distinct inputs have distinct outputs

Hence every $b \in B$ corresponds to *one and only one* $a \in A$, as required by the definition of function.

Invertibility: two notions not to be confused

- Here the inverse is defined on the **whole** codomain B: *both* injectivity *and* surjectivity are needed
- In the chapter "Inverse functions" we study functions of a real variable and a function is called *invertible* when it is only **injective**
- There the inverse is built on the **image** $f(D)$ and not on the whole codomain: with respect to the image surjectivity is automatic

Example: inverse of a bijection (1/2)

Let us go back to the bijection $f : \mathbb{N} \to \mathbb{Z}$,

$$f(n) = (-1)^n \left\lceil \frac{n}{2} \right\rceil,$$

which pairs $0 \leftrightarrow 0$, $1 \leftrightarrow -1$, $2 \leftrightarrow 1$, $3 \leftrightarrow -2$, $4 \leftrightarrow 2, \ldots$

Its inverse $f^{-1} : \mathbb{Z} \to \mathbb{N}$ is:

$$f^{-1}(m) = \begin{cases} 2\,m & \text{if} \quad m \geq 0 \\ -(2\,m+1) & \text{if} \quad m < 0 \end{cases}$$

Example: inverse of a bijection (2/2)

- If $m \geq 0$, then $n = 2\,m$ is even and

$$f(n) = (-1)^{2m}\lceil m \rceil = m$$

- If $m < 0$, then $n = -(2\,m + 1)$ is odd and nonnegative, and

$$f(n) = (-1)^n \left\lceil \frac{-2\,m - 1}{2} \right\rceil = -\left\lceil -m - \frac{1}{2} \right\rceil = -(-m) = m,$$

because $-m$ is a positive integer and hence $\left\lceil -m - \frac{1}{2} \right\rceil = -m$

Summary

Inverse of a bijection:

- If $f : A \to B$ is a bijection, $f^{-1} : B \to A$ is defined by $f^{-1}(b) = a \Leftrightarrow f(a) = b$
- Surjectivity gives the existence, injectivity the uniqueness of the input associated with each output
- $f^{-1}(f(a)) = a$ and $f(f^{-1}(b)) = b$

Final summary

- A **function** associates with each admissible input *one and only one* output
- $f : A \to B$: A is the **domain**, B the **codomain**, $f(A) \subseteq B$ the **image**
- Two functions are **equal** if they have the same domain, the same codomain and the same values
- Functions can be **surjective**, **injective** or **bijective**
- For **finite** sets: surjective $\Rightarrow |A| \geq |B|$, injective $\Rightarrow |A| \leq |B|$
- Every bijection $f : A \to B$ has an **inverse** $f^{-1} : B \to A$